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FEZrs: An Open-Source Python Package for Geospatial Multispectral Image Processing and Remote Sensing

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Summary

The history of remote sensing has evolved from early nineteenth-century balloon photography to modern multispectral and satellite-based Earth observation systems, enabling large-scale environmental monitoring and geospatial analysis workflows (Cracknell, 2018). Over time, remote sensing technologies have expanded from aerial observation platforms to advanced satellite (Zhang et al., 2022) and unmanned aerial vehicle (Simic Milas et al., 2018) systems that support scientific research, environmental assessment, and operational geospatial applications.

The increasing availability of multispectral remote sensing imagery from platforms such as Landsat (Williams et al., 2006) and Sentinel (Malenovský et al., 2012) has accelerated the adoption of spectral analysis methods across a broad range of remote sensing tasks and applications (Montero et al., 2023; Xue & Su, 2017). These developments have expanded remote sensing use in geoscience (Camps-Valls et al., 2021) and mineral resource assessment (Sabins, 1999). The same imagery now also supports environmental engineering, wildlife monitoring (Melesse et al., 2007; Shukla et al., 2024), biology, marine science (Chassot et al., 2011), and agriculture (Sishodia et al., 2020).

As remote sensing datasets continue to grow in scale and complexity, researchers and practitioners increasingly rely on accessible computational tools for image analysis, feature extraction, and spectral interpretation. These workflows often require standardized implementations of widely used spectral indices (Montero et al., 2023) and consistent processing utilities to ensure reproducibility and efficiency in both research and operational applications. FEZrs is designed to simplify remote sensing data processing and interpretation by providing a consistent framework for a wide range of analytical workflows.

Although primarily developed for geoscience applications, FEZrs can also be applied more broadly to multispectral and hyperspectral analysis in diverse fields beyond the geosciences.

Statement of Need

FEZrs is an open-source project for remote sensing image processing and geospatial analysis that provides a Python package for spectral analysis, image enhancement, filtering, spectral

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index computation, and feature extraction. The package integrates established remote sensing workflows within a unified open-source framework, supporting both research and industrial applications.

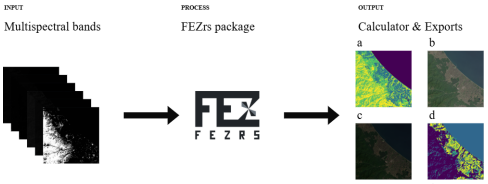


Figure 1: Overview of the FEZrs processing workflow. Multispectral remote sensing imagery is processed within the FEZrs framework to generate representative outputs, including (a) k-means clustering, (b) gamma-corrected RGB enhancement, (c) RGB image representation, and (d) Support Vector Machine (SVM) classification.

Modern remote sensing studies frequently rely on multiple analytical approaches. FEZrs implements a collection of widely used spectral indices (Montero et al., 2023), including the Normalized Difference Vegetation Index (NDVI) (Huang et al., 2021), Normalized Difference Water Index (NDWI) (McFeeters, 1996), Soil Adjusted Vegetation Index (SAVI) (Huete, 1988), Aerosol Free Vegetation Index (AFRI) (Karnieli et al., 2001), Bare Soil Index (BSI) (Rikimaru et al., 2002), and Urban Index (UI) (Deng & Wu, 2012). These indices are widely used for vegetation monitoring, water-resource assessment, environmental monitoring, land-surface characterization, and urban analysis.

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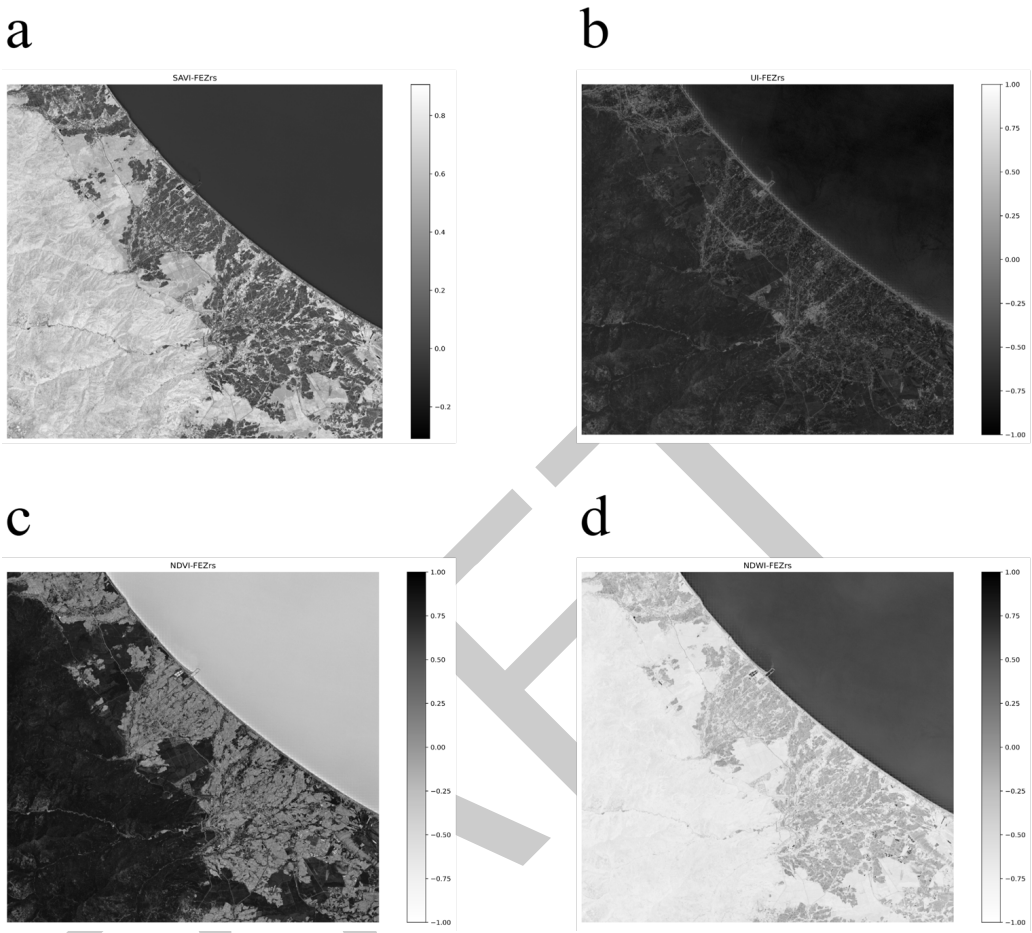


Figure 2: Representative spectral index outputs generated by FEZrs, including (a) Soil Adjusted Vegetation Index (SAVI), (b) Urban Index (UI), (c) Normalized Difference Vegetation Index (NDVI), and (d) Normalized Difference Water Index (NDWI). All outputs were computed from the same multispectral dataset using the corresponding FEZrs implementations.

The package also incorporates image enhancement and filtering techniques, including Gaussian smoothing (Lindeberg, 2024), median filtering (Justusson, 1981), Laplacian (Merris, 1994), and Sobel operators (Chang et al., 2023), which are widely used for noise reduction, edge detection, and image interpretation in remote sensing applications. Texture-based analysis is supported through the Gray-Level Co-occurrence Matrix (GLCM) (Utaminingrum et al., 2023), a widely adopted approach for characterizing spatial patterns in land-cover mapping and environmental studies. FEZrs further provides Principal Component Analysis (PCA) for dimensionality reduction and exploratory analysis of multispectral data (Jolliffe & Cadima, 2016), as well as k-means clustering for unsupervised image classification and pattern discovery (Sinaga & Yang, 2020). In addition, Support Vector Machine (SVM) classification is included as a supervised machine-learning approach for land-cover classification and feature discrimination in remote sensing datasets (Yue et al., 2003). Spectral profile analysis enables the investigation of spectral responses across wavelengths and supports multispectral interpretation workflows.

Change Detection	Filters	Image Enhancement	Spectral Indices	Additional Tools
Burn Indices	Gaussian Laplacian	Original Float	AFRI BI	KMeans GLCM

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Change Detection	Filters	Image Enhancement	Spectral Indices	Additional Tools
MagDir	Mean	Equalize	NDVI	HSV
SubDiv	Median	Adaptive	NDWI	IRHSV
Time	Sobel	Gamma	SAVI	Mosaic
		LogAdjust	UI	PCA
		SigmoidAdjust		SpectralProfile
		OriginalRGB		SVM
		EqualizeRGB		
		AdaptiveRGB		
		GammaRGB		

Table 1: Overview of the implemented FEZrs modules, which enable users to perform common remote sensing processing tasks, spectral analysis, feature extraction, and image-based analysis workflows within a single Python package.

Although these methods are widely utilized, there is still considerable value in open-source tools that bring together essential remote sensing functionalities within a single, coherent analytical platform. While FEZrs was developed with geoscience applications in mind, its functionality is equally applicable to multispectral and hyperspectral data analysis across a variety of scientific and technical domains.

State of the Field

FEZrs is built on general-purpose libraries (OpenCV (Xie & Lu, 2013), scikit-image (Walt et al., 2014), scikit-learn (Kramer, 2016), NumPy (Harris et al., 2020), matplotlib (Hunter, 2007), pandas (McKinney, 2010), Rasterio (Gillies & others, 2013), GDAL (Warmerdam, 2008), and GeoPandas (Jordahl et al., 2020)) rather than competing with them. The relevant comparison is with domain-level remote sensing packages that already implement overlapping methods.

The most similar tool is scikit-eo (Tarazona et al., 2024), a JOSS package that covers spectral indices, PCA, supervised classification including SVM, data fusion, and accuracy assessment. spyndex and the Awesome Spectral Indices catalogue (Montero et al., 2023) standardize more than two hundred index definitions; FEZrs implements a small, documented subset of those indices rather than a catalogue. Ndvi2Gif (García Díaz, 2026) provides multi-seasonal vegetation analysis on Google Earth Engine, a cloud time-series workflow that is complementary to FEZrs's local raster calculators. Other tools occupy adjacent parts of the same space: earthpy and EOReader for scene handling, rioxarray for labeled rasters, RSGISLib and Orfeo Toolbox for processing pipelines, and torchgeo for deep-learning workflows.

Those packages are capable, and FEZrs is not a replacement for any of them. What it adds is a uniform calculator API: thirty-seven heterogeneous operations (indices, filters, enhancement, GLCM, PCA, clustering, change detection, SVM) behind one three-line calling pattern, each capable of producing a publication-ready figure via `execute()` and a georeferenced GeoTIFF via `to_raster()` without plotting or I/O code from the user. That uniformity is aimed at teaching and rapid exploratory work on multispectral scenes, where assembling a bespoke toolchain for each step is the usual cost.

Package	Scope	What FEZrs adds
scikit-eo	Spectral indices, PCA, SVM, fusion, accuracy assessment	Same methods behind one calculator interface, plus publication figures

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Package	Scope	What FEZrs adds
spyndex / Awesome Spectral Indices	Catalogue of 200+ index definitions	A small teaching set with `execute()` / `to_raster()` export
Ndvi2Gif	Multi-seasonal vegetation analysis on Google Earth Engine	Complementary cloud time-series workflows; FEZrs operates on local rasters
Rasterio, GDAL, OpenCV, scikit-image, scikit-learn	I/O, image processing, and ML building blocks	Domain workflows layered on top of these libraries
earthpy, EOReader, rioxarray, RSGISLib, Orfeo Toolbox, torchgeo	Scene I/O, preprocessing pipelines, deep learning	Complementary; FEZrs does not replace them

Table 2: Domain-level packages with overlapping scope, and the narrower contribution FEZrs makes relative to each.

Software design

FEZrs is organized around a shared abstract base class (`BaseTool`) that standardizes the lifecycle of every analysis module: input validation, processing, visualization customization, and export. Each calculator (for example, `NDVICalculator`, `GaussianCalculator`, or `KMeansCalculator`) inherits this interface, accepts band file paths through a common file-handling layer, and exposes a small set of methods such as `execute` and `histogram_export`. This pattern keeps the public API uniform across spectral indices, filters, enhancement tools, change-detection utilities, and machine-learning modules, reducing cognitive overhead when composing multi-step workflows.

The package is modular by domain (`spectral_indices`, `filters`, `image_enhancement`, `change_detection`, `clustering`, `glcm`, `pca`, `svm`, and others), while shared utilities handle band-path typing, file I/O, and histogram support. Raster inputs are typically multi-band geospatial imagery. Results can be exported two ways: as rendered figures via `execute()`, for inspection and reporting, and as georeferenced GeoTIFF rasters via `to_raster()`, which preserve full numerical precision along with the coordinate reference system and affine transform of the source imagery, and are therefore suitable for GIS overlay, zonal statistics, and multitemporal analysis. The design deliberately favors composition of independent calculators over a single monolithic pipeline object, so researchers can select only the methods required for a given study while still benefiting from consistent validation and export behavior.

Research Impact Statement

Since its release on PyPI in February 2025, FEZrs has received more than 100,000 downloads according to PePy, indicating substantial distribution of the package, with download statistics potentially including traffic from PyPI mirrors. FEZrs is openly available through GitHub, PyPI, and the Anaconda ecosystem as an open-source Python package.

FEZrs has been employed in research on remote sensing and geospatial analysis. It has been used for principal component analysis (PCA) of Landsat 9 multispectral imagery (Farmahini Farahani et al., 2024), comparison of k-means clustering and spectral indices for land use and land cover characterization (Rezaei & Farmahini Farahani, 2024), and a comparative assessment of SVM, k-means, and spectral indices for land cover classification in engineering geology (Talkhablou et al., 2025). These published applications were conducted by members

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of the FEZrs development team and demonstrate the software's use in practical research workflows. FEZrs has also supported geospatial analysis of the Torud Playa area in Isfahan, Iran, providing an additional example of its application in field-based remote sensing and geospatial workflows. In addition, FEZrs received the Merit Award (Ideator Category) at the 9th Iran National Young Scientists Festival.

The project provides comprehensive documentation, practical examples, a test suite, and a modular architecture to support reproducible research and operational geospatial workflows. Detailed installation instructions for both pip (`pip install fezrs`) and conda (`conda install -c FEZtool fezrs`) are available in the FEZrs Installation Guide.

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The authors would like to express their sincere appreciation to the global open-source developer community. The collective efforts, shared knowledge, and freely available tools developed by this community have provided the essential foundation that made this project possible. Their commitment to collaboration, transparency, and innovation continues to advance technological progress and enables researchers to build upon and improve existing work. The authors gratefully acknowledge these invaluable contributions.

AI Usage Disclosure

While the core software design, implementation, and development decisions of FEZrs were made by the authors, AI-assisted tools were used to assist with unit testing, validation workflows, and manuscript refinement. All AI-assisted outputs were independently reviewed, verified, and validated by the authors.

Conflict of interest

The authors declare no conflict of interest.

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